

Projects for Computational Physics

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Monte Carlo Optimisation: The traveling salesman problem

Preparation(*), Implementation(**), Analysis(*)

Project Description

The traveling salesman problem (TSP) is a long-standing and often-discussed issue in computational physics and operations research: a traveling (owing to political correctness) salesperson has to find the shortest closed tour between a certain set of cities, where each city is only visited once during the tour. The problem can for instance be tackled quite efficiently using Monte Carlo algorithms.

Naturally, this problem has applications in many every-day problems, e.g. in finding directions using a route guiding system. Optimisation can be achieved with respect to different functions, e.g. the shortest path, the quickest path, and others.

2. Tasks / hints / interesting problems¹

- Start with a simple problem: a few points distributed uniformly on a circle. Solve the problem using crude combinatorical methods (trial-and-error).
- Implement a Monte Carlo-algorithm to solve the problem approximately (see paper [1]). Use a random path to start with (Using 1-2-3-... on a circle does not really help here!). Compare the efficiency of both methods.
- Use more points. Once you reach the equilibrium distribution for a certain “temperature“ adjust it and study the effect on the acceptance rate by plotting traveling distance against Monte Carlo trials.
- Study different setups: points distributed uniformly on a square, two separated squares (see paper [1]), or whatever you can think of.
- Study the time-dependent traveling salesman problem (TDTSP), that is include traffic jams at rush hours, define inner city parts, etc. (see also paper [2])

Literature

- [1] “Thermodynamical approach to the traveling salesman problem: An efficient simulation algorithm”, V. Cerny, JOTA, Vol 45, No 1, 1985
- [2] “Optimization of the time-dependent traveling salesman problem with Monte-Carlo methods”, Bentner et al., Phys.Rev. E, 64, 036701

¹These tasks should rather be considered a general guideline than a strict requirement. Besides, the list is by no means complete and could easily be extended

- [3] “The traveling salesman problem: A computational study”, D.L. Applegate, R.E. Bixby, V. Chvatal, W.J. Cook, Princeton University Press, 2007